# EXP7

April 10, 2026

[13]:

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split, learning_curve
from sklearn.metrics import accuracy_score, classification_report
from sklearn.datasets import make_classification
```

[15]:

```python
X, y = make_classification(
    n_samples=5000,
    n_features=10,
    n_informative=6,
    n_redundant=2,
    random_state=42
)

feature_names = [f'Feature_{i}' for i in range(X.shape[1])]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

[17]:

```python
rf = RandomForestClassifier(
    n_estimators=100,
    oob_score=True,
    random_state=42,
    n_jobs=-1
)

rf.fit(X_train, y_train)

print("OOB Score:", rf.oob_score_)
```

```
OOB Score: 0.912
```

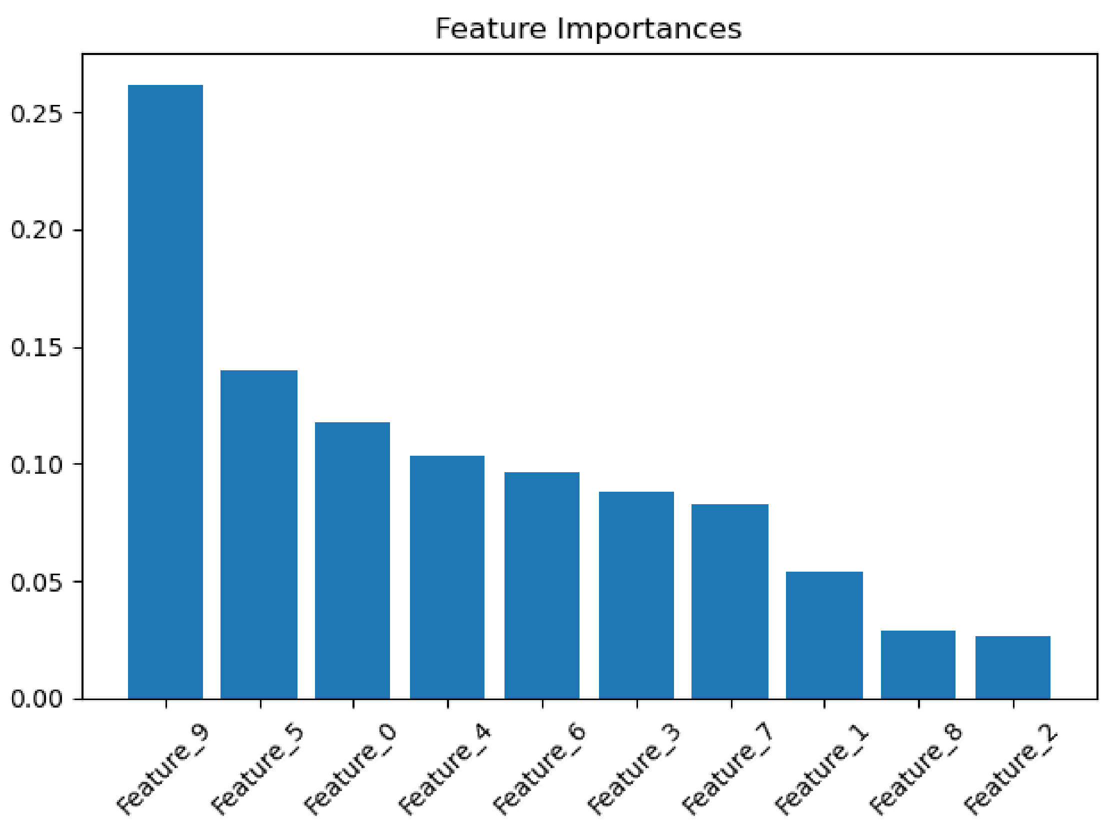
[19]:

```python
y_pred = rf.predict(X_test)
```

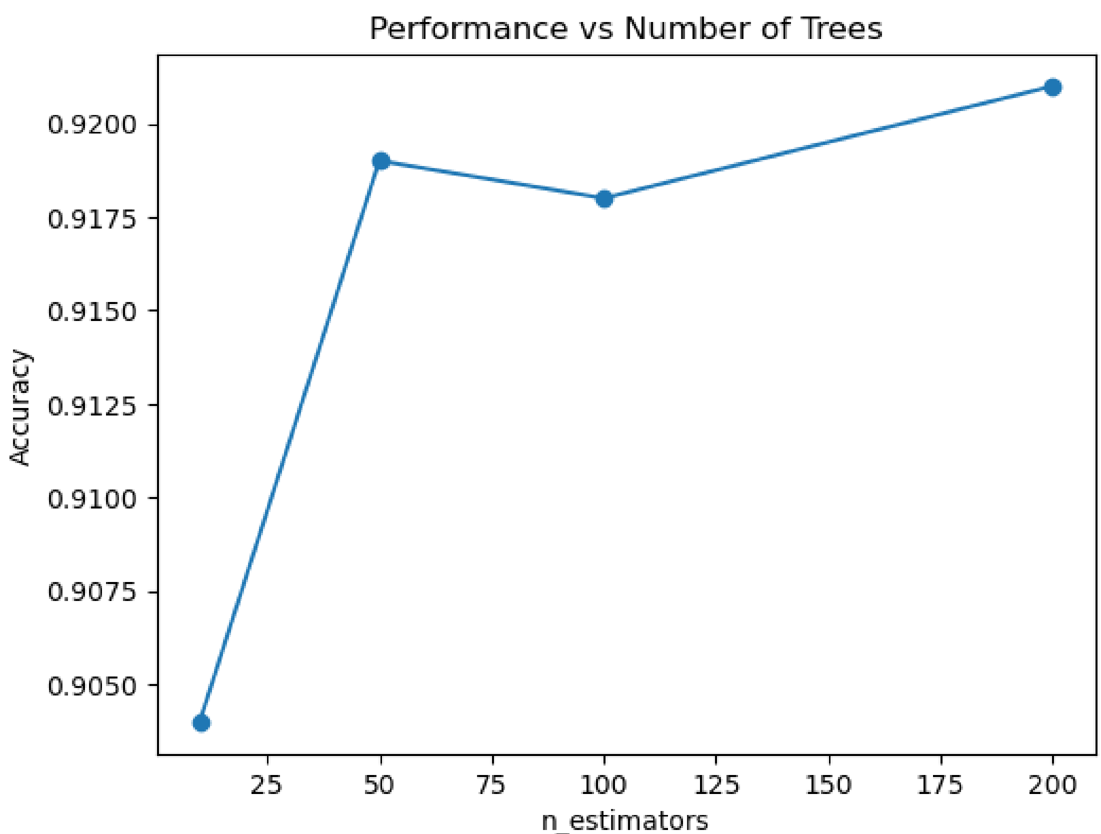
```python
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.918
              precision    recall  f1-score   support

           0       0.92      0.92      0.92       504
           1       0.92      0.91      0.92       496

    accuracy                           0.92      1000
   macro avg       0.92      0.92      0.92      1000
weighted avg       0.92      0.92      0.92      1000
```

[21]:
```python
importances = rf.feature_importances_

indices = np.argsort(importances)[::-1]

plt.figure()
plt.title("Feature Importances")
plt.bar(range(X.shape[1]), importances[indices])
plt.xticks(range(X.shape[1]), [feature_names[i] for i in indices], rotation=45)
plt.tight_layout()
plt.show()
```

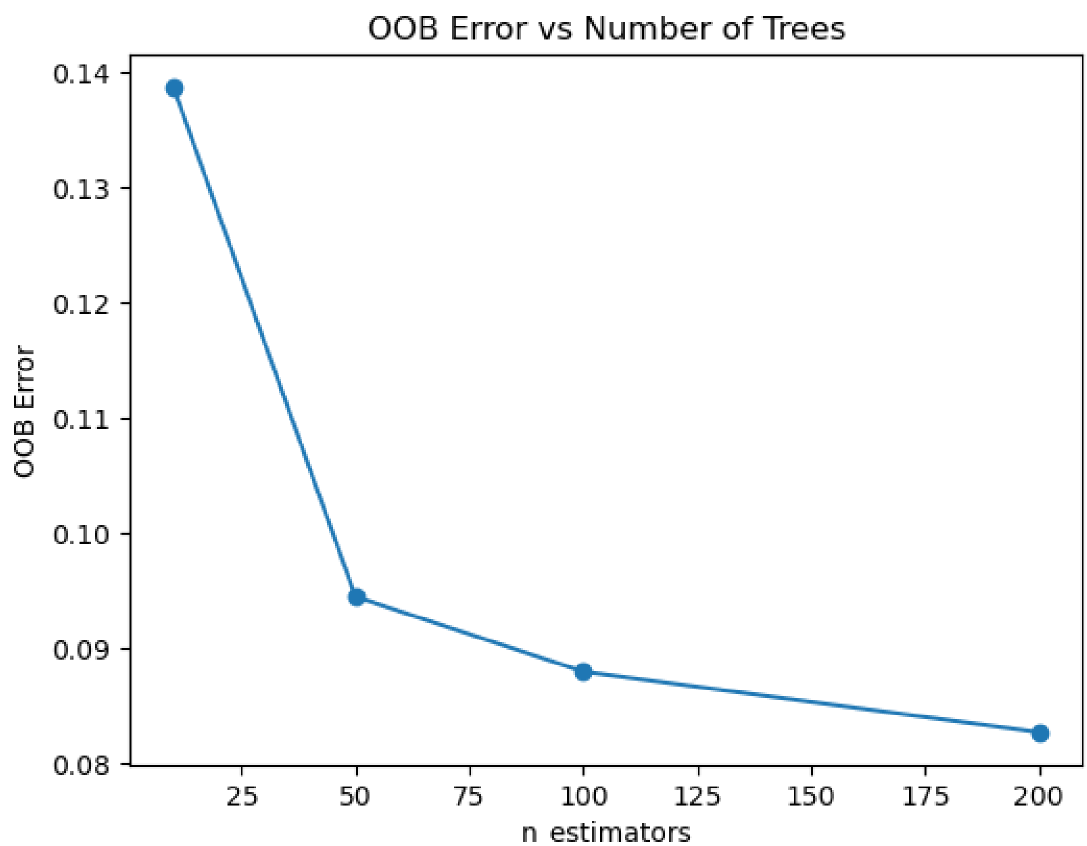


[23]:
```
estimators = [10, 50, 100, 200]
accuracies = []

for n in estimators:
    model = RandomForestClassifier(n_estimators=n, random_state=42)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracies.append(accuracy_score(y_test, y_pred))

plt.figure()
plt.plot(estimators, accuracies, marker='o')
plt.xlabel("n_estimators")
plt.ylabel("Accuracy")
plt.title("Performance vs Number of Trees")
plt.show()
```



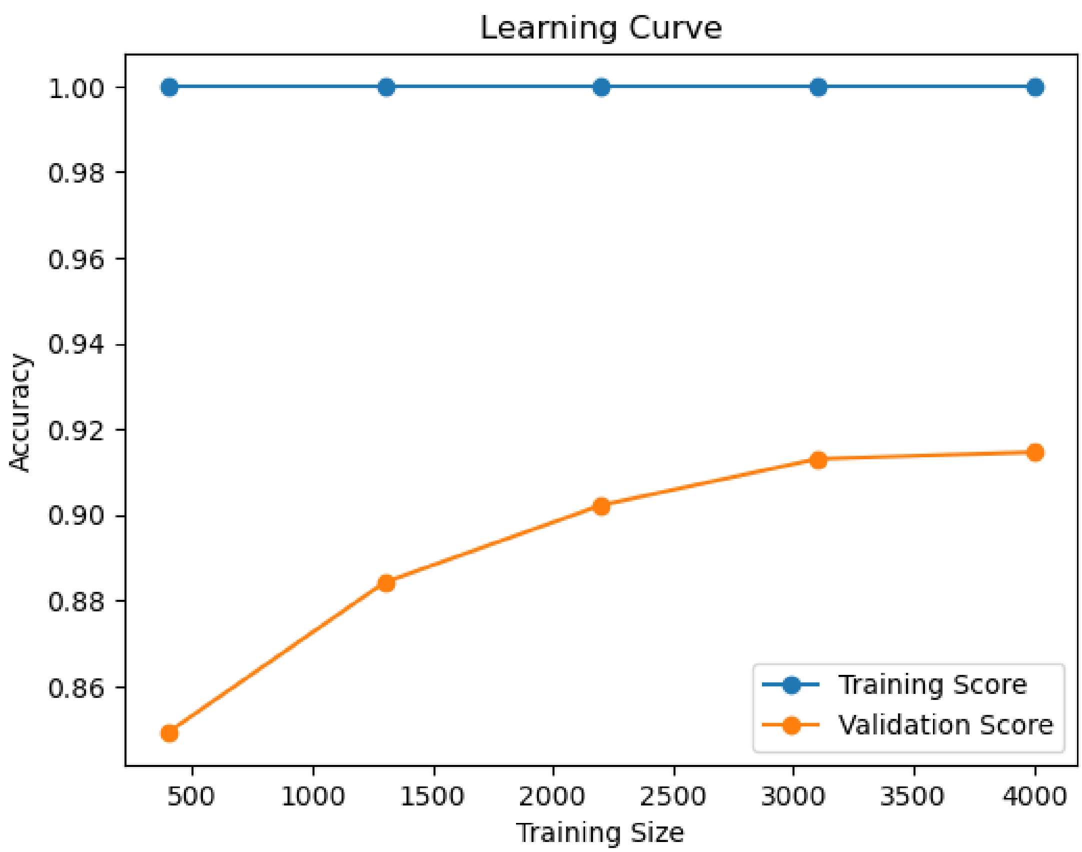
[25]:
```
oob_errors = []

for n in estimators:
    model = RandomForestClassifier(
        n_estimators=n,
        oob_score=True,
        bootstrap=True,
        random_state=42
    )
    model.fit(X_train, y_train)
    oob_errors.append(1 - model.oob_score_)

plt.figure()
plt.plot(estimators, oob_errors, marker='o')
plt.xlabel("n_estimators")
plt.ylabel("OOB Error")
plt.title("OOB Error vs Number of Trees")
plt.show()
```

```
/opt/anaconda3/lib/python3.12/site-packages/sklearn/ensemble/_forest.py:615:
UserWarning: Some inputs do not have OOB scores. This probably means too few
trees were used to compute any reliable OOB estimates.
  warn(
```



[27]:

```
train_sizes, train_scores, test_scores = learning_curve(
    RandomForestClassifier(n_estimators=100, random_state=42),
    X, y,
    cv=5,
    scoring='accuracy',
    n_jobs=-1
)

train_mean = np.mean(train_scores, axis=1)
test_mean = np.mean(test_scores, axis=1)

plt.figure()
plt.plot(train_sizes, train_mean, marker='o', label="Training Score")
plt.plot(train_sizes, test_mean, marker='o', label="Validation Score")

plt.xlabel("Training Size")
plt.ylabel("Accuracy")
plt.title("Learning Curve")
plt.legend()
plt.show()
```



```
[ ]:
```